

Lecture 4: Randomized Quick Sort

*Instructor: Sourav Chakraborty, Sanjukta Roy**Scribe: Rohan, Saksham*

1 Motivation

In deterministic quick sort, choosing a pivot affects the time complexity

$$\mathcal{T}(n) = \mathcal{T}(n-1) + \mathcal{O}(n) \quad \text{When choosing a bad pivot}$$

$$\mathcal{T}(n) = 2\mathcal{T}\left(\frac{n}{2}\right) + \mathcal{O}(n) \quad \text{When choosing a good pivot}$$

Time complexities for these cases:

$$\text{Worst case : } \mathcal{O}(n^2)$$

$$\text{Best Case : } \mathcal{O}(n \log n)$$

2 Randomized Quick Sort

Randomized Quick Sort introduces randomness or some sort of probabilistic method to choose the pivot. Choosing different pivots, affects the **number of comparisons** in the next iteration in the recursion, which in turn affects the **time complexity**.

2.1 Choosing Pivot

One way to choose a pivot is to flip $\log n$ many coins. And write 1 for head and 0 for tail. This will give a binary string of length $\log n$. Converting this binary string into decimal number, we have a number between 1 and n . Then we choose the corresponding element at that position in the array as the pivot.

Another way is to randomly pick an element of the array uniformly at random as pivot.

Las Vegas Algorithm :

A **Las Vegas Algorithm** is a class of randomized algorithms characterized by the following property :

- Keep running the algorithm until we get a correct output.

The running time is a random variable whose value depends on the algorithm's internal random choices. So, we speak of **Expected Time Complexity** here.

Monte Carlo Algorithm :

A **Monte Carlo Algorithm** is a class of randomized algorithms characterized by the following property :

- Repeat the algorithm a bounded number of times, with a guarantee of correct output with '**high**' probability.

Here 'high' is a convention. Generally it means that the probability of failure is something inverse polynomial in input size.

Here, we have a deterministic running time and that is bounded.

2.2 Algorithm (RQS)

We choose the pivot uniformly at random from the input array.

- S = List of numbers
- p = Chosen pivot uniformly at random

Then we split the array into two parts around the pivot such that.

$$S_1 = \{y \mid y \in S, y < p\}$$

$$S_2 = \{y \mid y \in S, y > p\}$$

This is the only step where the comparison is happening.

(In quick sort comparison happens while breaking the array and in merge sort comparison happens while merging the array.)



Then we recursively solve S_1 and S_2 .

2.3 Time Complexity Calculation

We count the number of comparisons which depends on the random choice of the pivot.

So, counting the number of comparisons is our random variable.

Let, X = Number of comparisons.

We need to find the expected number of comparisons. i.e. $\mathbb{E}(X)$ Let,

$$X_{ij} = \begin{cases} 1, & \text{if } a_i \text{ and } a_j \text{ are comparable} \\ 0, & \text{otherwise} \end{cases}$$

Now,

$$\begin{aligned} X &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n X_{ij} \\ \implies \mathbb{E}[X] &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n \mathbb{E}[X_{ij}] \quad (\text{by linearity of expectation}) \end{aligned}$$

Now,

$$\begin{aligned} \mathbb{E}[X_{ij}] &= 1 \times \mathbb{P}(X_{ij} = 1) + 0 \times \mathbb{P}(X_{ij} = 0) \\ \implies \mathbb{E}[X_{ij}] &= \mathbb{P}(X_{ij} = 1) \\ \implies \mathbb{E}[X] &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n \mathbb{P}(X_{ij} = 1) \end{aligned}$$

Observe, in the sorted array $B = \{b_1, b_2, \dots, b_{i-1}, b_i, \dots, b_j, b_{j+1}, \dots, b_n\}$ If an element outside $B_{ij} = \{b_i, \dots, b_j\}$ is chosen as the pivot element then b_i, b_j will lie in the same side of the subarray in the next step.

If an element from $B_{ij} = \{b_i, \dots, b_j\}$ is chosen other than b_i or b_j , then these two elements will be on the opposite side of the pivot. Therefore, in the subsequent subarray they can not be compared anymore.

$$\implies X_{ij} = 1 \iff b_i \text{ or } b_j \text{ is chosen as the pivot element at any step}$$

So, we need to choose b_i or b_j before $\{b_{i+1}, \dots, b_{j-1}\}$ elements for the pivot, otherwise b_i and b_j will lie in different arrays. Now, $|B_{ij}| = j - i + 1$ and,

$$\begin{aligned} \mathbb{P}(b_k \text{ is chosen first as pivot element from the set } A_k) &= \frac{1}{|B_{ij}|} \\ \implies \mathbb{P}(b_i \text{ or } b_j \text{ being chosen as pivot element from the set } B_{ij}) &= \frac{2}{|B_{ij}|} = \frac{2}{j - i + 1} \end{aligned}$$

Therefore,

$$\mathbb{P}(X_{ij} = 1) = \frac{2}{j - i + 1} \implies \mathbb{E}[X_{ij}] = \frac{2}{j - i + 1}$$

So, expected number of comparisons:

$$\begin{aligned} \mathbb{E}[X] &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n \mathbb{E}[X_{ij}] = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{2}{j - i + 1} \\ \implies \mathbb{E}[X] &= \sum_{i=1}^{n-1} \sum_{k=2}^{n-i+1} \frac{2}{k} \quad (\text{Substitute } j - i + 1 = k) \\ \implies \mathbb{E}[X] &\leq 2 \sum_{i=1}^{n-1} \sum_{k=1}^n \frac{1}{k} \leq 2 \sum_{i=1}^n c \cdot \log_2 n \\ \implies \mathbb{E}[X] &\leq 2.n.c. \log_2 n \\ \implies \mathbb{E}[X] &= \mathcal{O}(n \cdot \log n) \end{aligned}$$

Therefore, the expected number of operations $\mathcal{O}(n \log n)$

So, Expected time complexity of Randomized Quick sort is :

$$\mathcal{O}(n \log n)$$

3 Markov's Inequality

Statement: For a non-negative random variable X and a constant $a > 0$, the following holds:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

where, $E[.]$ is the expected value.

Proof: Let I be a random variable defined as follows:

$$I = \begin{cases} 1 & , \text{ if } X \geq a \\ 0 & , \text{ o/w} \end{cases}$$

then one can observe that $I \leq \frac{X}{a}$ for all values of X .

$$\implies E[I] \leq E\left[\frac{X}{a}\right]$$

$$\implies P(X \geq a) \leq \frac{E[X]}{a} \quad \square$$

Problem 1. What is the probability of getting heads at least $\frac{3n}{4}$ times if a fair coin is tossed n times?

Solution: Let X be the number of heads obtained in the experiment of tossing a fair coin n number of times. Then, $X \sim \text{Bin}\left(n, \frac{1}{2}\right)$. Then using Markov's inequality we get,

$$\begin{aligned} P\left(X \geq \frac{3n}{4}\right) &\leq \frac{E[X]}{(3n/4)} \\ &= \frac{n/2}{3n/4} \quad (\text{If } X \sim \text{Bin}(n, p), \text{ then } E[X] = np) \\ \implies P\left(X \geq \frac{3n}{4}\right) &\leq \frac{2}{3} \end{aligned}$$